| Setting | Method | Asia | | Sachs | | Alarm | | Insurance | | Hailfinder | | SIIM-Pn. | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | T | C. Cov. | T | C. Cov. | T | C. Cov. | T | C. Cov. | T | C. Cov. | T | C.Cov. |
| Cent. (Upper Bound) | OpaqNN | 80.6±2.0 | 100 | 77.5±2.1 | 100 | 73.6±1.5 | 100 | 76.6±2.0 | 100 | 73.2±0.8 | 100 | 75.5±0.8 | 100 |
| | CBM | 79.9±2.0 | 100 | 77.2±1.7 | 100 | 73.7±1.2 | 100 | 76.4±2.0 | 100 | 75.8±1.5 | 100 | 75.1±0.6 | 100 |
| | CEM | 80.6±2.0 | 100 | 77.6±1.7 | 100 | 70.7±0.6 | 100 | 76.2±2.2 | 100 | 72.6±1.9 | 100 | 75.1±0.4 | 100 |
| | CGM | 79.3±2.2 | 100 | 76.5±1.7 | 100 | 70.4±1.2 | 100 | 73.2±1.9 | 100 | 72.4±1.8 | 100 | 69.5±2.2 | 100 |
| | $C^2$BM | 80.1±2.0 | 100 | 77.4±1.8 | 100 | 72.0±1.2 | 100 | 75.1±2.0 | 100 | 74.1±1.8 | 100 | 74.4±0.6 | 100 |
| Loc. | OpaqNN | 62.9±8.8 | 43.3±4.1 | 55.1±16.1 | 56±4.0 | 36.1±18.2 | 40.9±7.1 | 19.2 ± 12.3 | 70.5±7.4 | 44.7±15.2 | 63.7±3.5 | 61.5±4.7 | 37.5±0.4 |
| | CBM | 48.7±6.4 | 43.3±4.1 | 45.9±20.9 | 56±4.0 | 45.9±11.0 | 40.9±7.1 | 31.2±15.7 | 70.5±7.4 | 45-4±13.3 | 63.7±3.5 | 59.2±11.1 | 37.5±0.4 |
| | CEM | 65.8±15.0 | 43.3±4.1 | 57.3±14.4 | 56±4.0 | 36.6±15.8 | 40.9±7.1 | 39.0±10.8 | 70.5±7.4 | 43.3±17.8 | 63.7±3.5 | 65.2±2.1 | 37.5±0.4 |
| | CGM | 67.4±8.4 | 43.3±4.1 | 50.5±16.0 | 56±4.0 | 36.6±17.5 | 40.9±7.1 | 32.1±14.1 | 70.5±7.4 | 38.1±8.8 | 63.7±3.5 | 63.1±0.1 | 37.5±0.4 |
| | $C^2$BM | 61.9±19.2 | 43.3±4.1 | 43.2±24.6 | 56±4.0 | 36.1±17.9 | 40.9±7.1 | 22.2± 23.3 | 70.5±7.4 | 36.7±7.8 | 63.7±3.5 | 63.1±0.1 | 37.5±0.4 |
| External Static FL | Fcl | 60.1±15.1 | N/A | 63.2±2.2 | N/A | 42.3±5.0 | N/A | 62.2±3.5 | N/A | 31.5±5.3 | N/A | 60.2±3.7 | N/A |
| | FedCBM | 63.2±11.0 | 43.3±8.0 | 57.3±19.4 | 56.0±7.8 | 46.9±16.5 | 68.7±11.9 | 59.7±17.1 | 83.2±5.1 | 49.9±13.3 | 84.7±5.5 | 64.9±7.3 | 65.0±9.2 |
| S-F-CMs | OpaqNN | 71.6±16.2 | 43.3±8 | 59.8±19.3 | 56±7.8 | 49.3±20.5 | 68.7±11.9 | 64.9±14.7 | 83.2±5,1 | 50.7±19.9 | 84.7±5.5 | 65.1±2.5 | 65±9.2 |
| | CBM | 59.9±11.0 | 43.3±8 | 50.9±26.1 | 56±7.8 | 41.3±21.0 | 68.7±11.9 | 58.5±23.3 | 83.2±5.1 | 57.3±17.9 | 84.7±5.5 | 62.9±12.8 | 65±9.2 |
| | CEM | 68.1±14.7 | 43.3±8 | 55.0±26.2 | 56±7.8 | 44.1±23.1 | 68.7±11.9 | 68.8±6.4 | 83.2±5.1 | 51.8±21.2 | 84.7±5.5 | 64.7±2.4 | 65±9.2 |
| | CGM | 67.1±14.4 | 43.3±8 | 44.0±28.8 | 56±7.8 | 52.6±18.5 | 48.7±20.6 | 57.9±23.0 | 76.8±11.6 | 42.9±14.8 | 84.7±5.5 | 63.2±0.2 | 65±9.2 |
| | $C^2$BM | 59.7±20.2 | 43.3±8 | 59.8±26.5 | 56±7.8 | 43.5±23.1 | 48.7±20.6 | 63.0±21.5 | 76.8±11.6 | 54.5±21.8 | 84.7±5.5 | 62.7±12.7 | 65±9.2 |
| **F-CMs** | OpaqNN | **80.0±1.8** | **100** | **75.1±2.1** | **100** | **72.9± 0.5** | **100** | **71.8±2.3** | **100** | **66.0±3.4** | **100** | **67.1±2.4** | **100** |
| | CBM | **79.8±1.4** | **100** | **75.3± 2.0** | **100** | **71.5±3.8** | **100** | **69.5± 1.1** | **100** | **72.4±3.6** | **100** | **71.0**±0.2 | **100** |
| | CEM | **80.5±1.7** | **100** | **76.0±2.6** | **100** | **73.4±0.4** | **100** | **71.0± 0.8** | **100** | **70.2±3.8** | **100** | **67.3±2.7** | **100** |
| | CGM | **80.0±2.0** | **100** | **70.2±3.2** | **100** | **69.1±5.9** | **100** | **70.8± 2.3** | **100** | **55.5±2.9** | **99.5±0.5** | **66.1±1.9** | **100** |
| | $C^2$BM | **80.4± 2.2** | **100** | **76.2±2.1** | **100** | **73.1±0.8** | **100** | **74.3± 2.3** | **100** | **72.8±2.1** | **99.5±0.5** | **68.8±2.5** | **100** |

Table 1: **Extended Table 1 of the manuscript, including additional baselines (Fcl, FedCBM).** *Task accuracy* (%) and *concept coverage* (%) with respect to the ground-truth concepts causally relevant for the task. Methods that improve the average performance are shown in bold.

| Data | Model | d_inst. | d_par. | agg. | n_rounds | tot. |
|---|---|---|---|---|---|---|
| Asia | F-CMs (CEM) | 0.227 ± 0.001 | 0.002 ± 0.000 | – | 42.583 ± 2.256 | 229.726 ± 13.609 |
| SIIM | F-CMs (CEM) | 0.691 ± 0.093 | 0.003 ± 0.000 | – | 35.350 ± 5.188 | 506.230 ± 75.622 |
| Asia | F-CMs (C2BM) | 0.244 ± 0.001 | 0.003 ± 0.000 | 0.063 ± 0.051 | 61.500 ± 3.679 | 370.800 ± 24.924 |
| SIIM | F-CMs (C2BM) | 0.672 ± 0.102 | 0.004 ± 0.000 | 0.037 ± 0.003 | 56.550 ± 7.278 | 922.543 ± 120.707 |

Table 2: **Calculation of average time (in seconds) for the different steps of our methodology**, reported as mean ±95% confidence interval over three runs. Specifically, *d_inst*, the time spent instantiating the architecture after client drift; *d_par*, the time spent initializing the post-drift architecture with the old learned parameters; *agg*, the time spent aggregating client graphs; *n_rounds*, the number of training rounds; and *tot.*, the total training time. These results are consistent with our complexity analysis, showing that the additional overhead introduced by F-CMs is modest and primarily due to lightweight, event-triggered operations, remaining small compared to the dominant cost of model training.

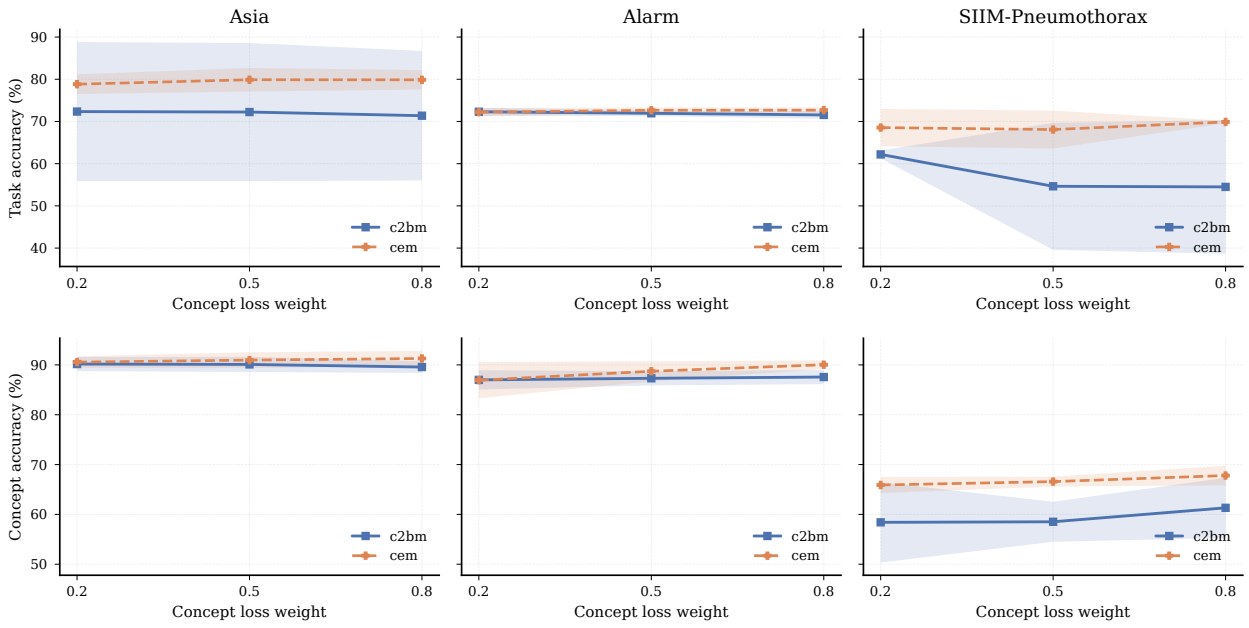


Figure 1: **Sensitivity analysis on $\gamma$.** *Task accuracy* (%) and *concept accuracy* (%) as a function of the concept loss weight ($\gamma$) for the C2BM and CEM instantiations across the Asia, Alarm, and SIIM-Pneumothorax datasets, calculated over three runs. The results show that downstream performance remains robust to changes in the concept-task trade-off, even on datasets with a larger number of concepts. A slight upward trend is observed in concept accuracy on SIIM-Pneumothorax as $\gamma$ increases, which is expected since the model places greater weight on the concept prediction loss, accompanied by a corresponding slight downtrend in task accuracy.

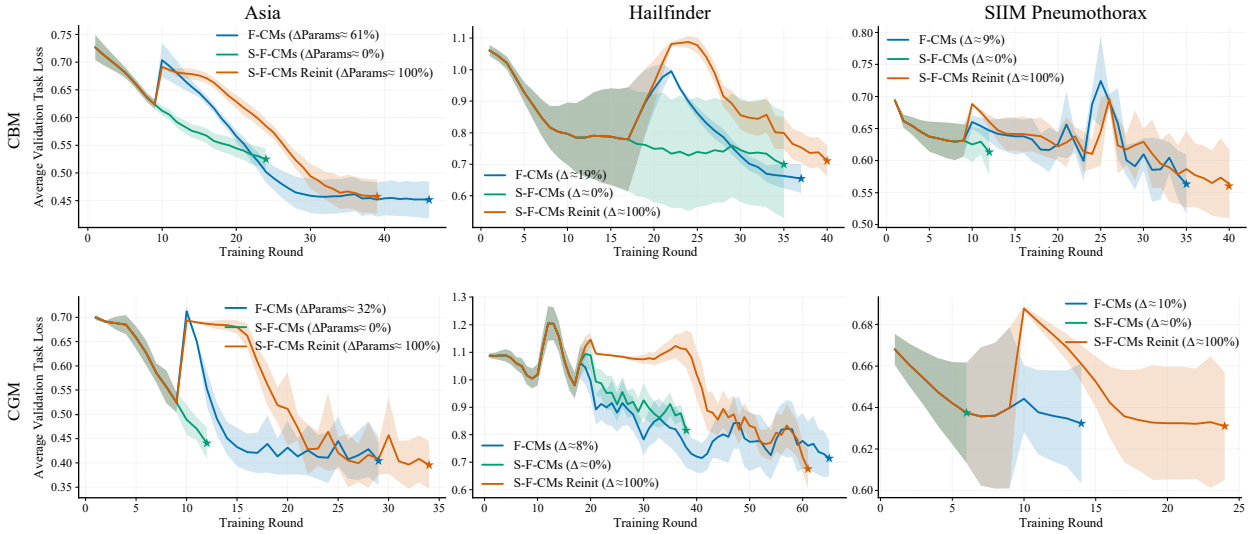


Figure 2: **Additional convergence results under temporal non-stationarity for CBM and CGM instantiations.** *Average validation task loss* across rounds, where new clients introduce unseen concepts. F-CMs (ours) adapt the architecture modularly and generally converge faster and/or to a lower loss than S-F-CMs (no adaptation) and S-F-CMs Reinit (full retraining), with gains maintained in most settings, though slightly smaller than in Fig. 3 of the original manuscript. Notably, our method always converges to lower average losses than S-F-CMs without adaptation and shows faster recovery than S-F-CMs Reinit after model adaptation (rounds 10 and 20). The legend reports the fraction of modified parameters.

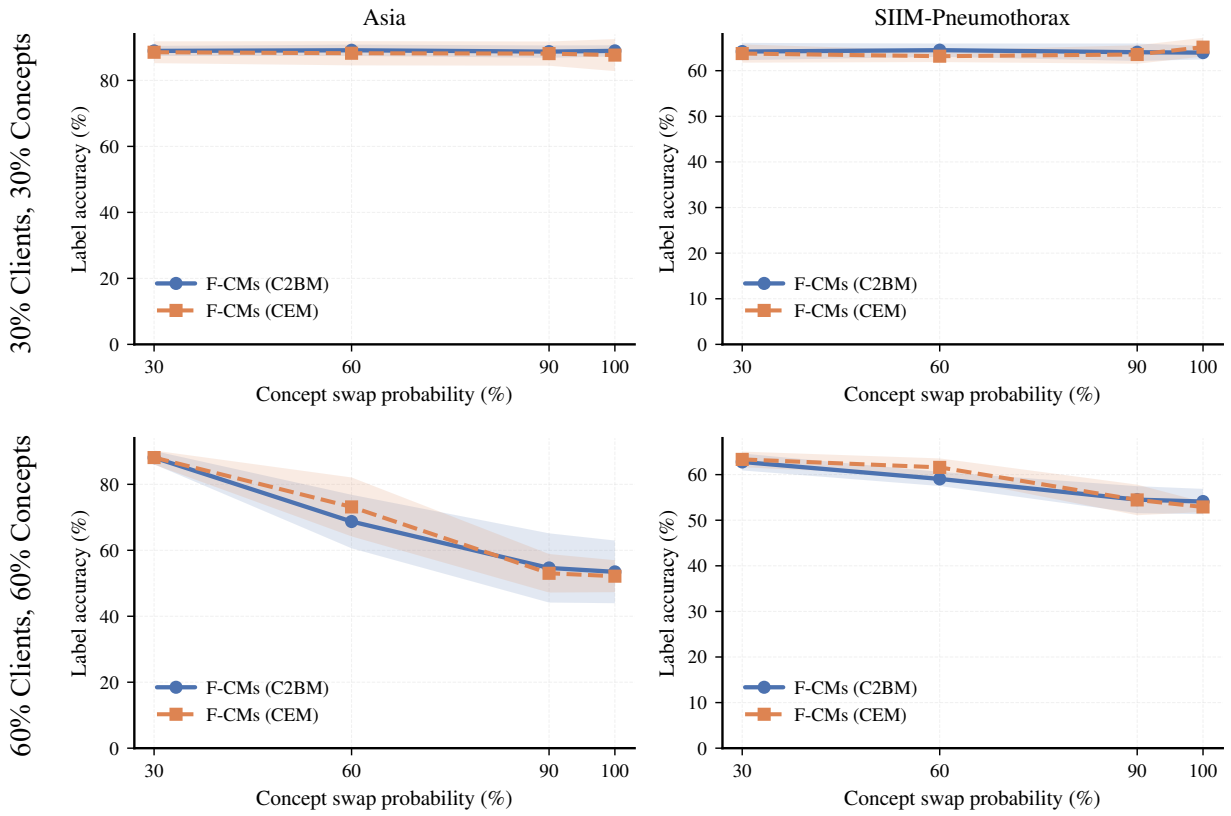


Figure 3: **Ablation study on inter-client semantic disagreement.** *Label accuracy (%)* of two representative F-CM instantiations, bipartite (F-CMs (CEM)) and graph-based (F-CMs (C2BM)), under varying fractions of concept label swaps within clients during training. Each row shows a configuration with either 30% of concepts and 30% of clients, or 60% of concepts and 60% of clients selected for swapping. For each concept, the corresponding fraction of clients is selected among those with supervision on it, and within them a fraction of the concept labels (x-axis: 30%, 60%, 90%, 100%) is swapped, simulating conflicting definitions across clients. Label accuracy is evaluated on a test set where concepts are not swapped; values are reported over three runs. Each column corresponds to a different dataset.

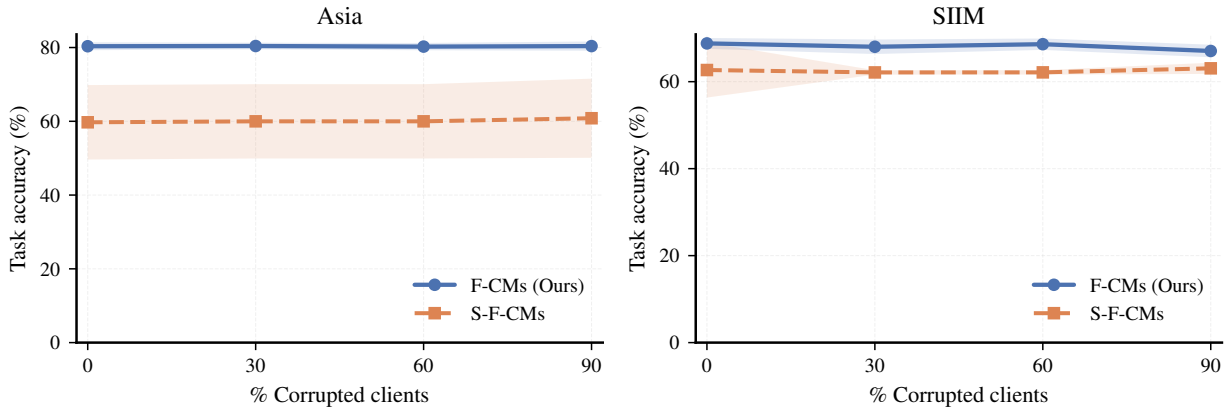


Figure 4: **Ablation study on downstream task robustness to graph misspecification.** *Task accuracy* of C2BM instantiations under the same perturbation setting as Figure 9, with graph alteration $p = 0.6$, shown as a function of the percentage of corrupted clients. On both Asia and SIIM-Pneumothorax, accuracy remains largely stable despite structural degradation, suggesting limited downstream sensitivity to moderate errors in the aggregated graph, while F-CMs consistently operates at a higher accuracy level than S-F-CMs.

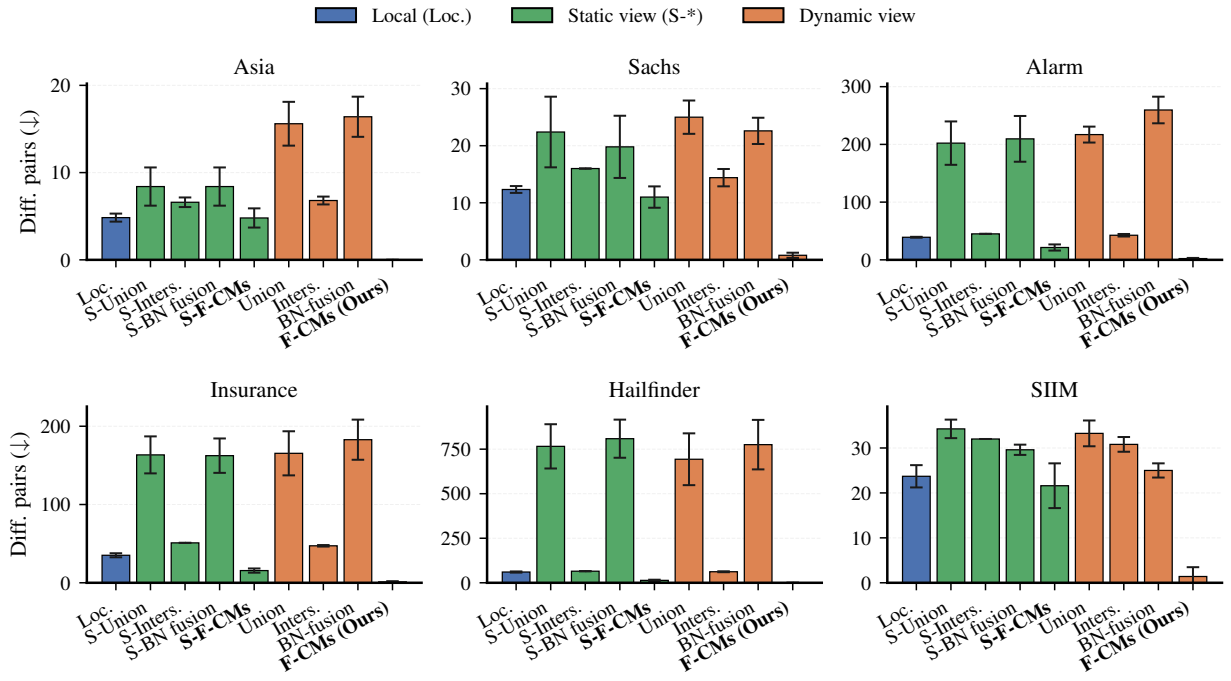


Figure 5: **Ablation of graph aggregation strategies under moderate perturbations.** *DiffPairs* (lower is better) for different aggregation rules under localized, static federated, and dynamic federated visibility, with graph alteration $p = 0.3$ and 30% corrupted clients, following the same experimental setup as Fig. 9 of the original manuscript.

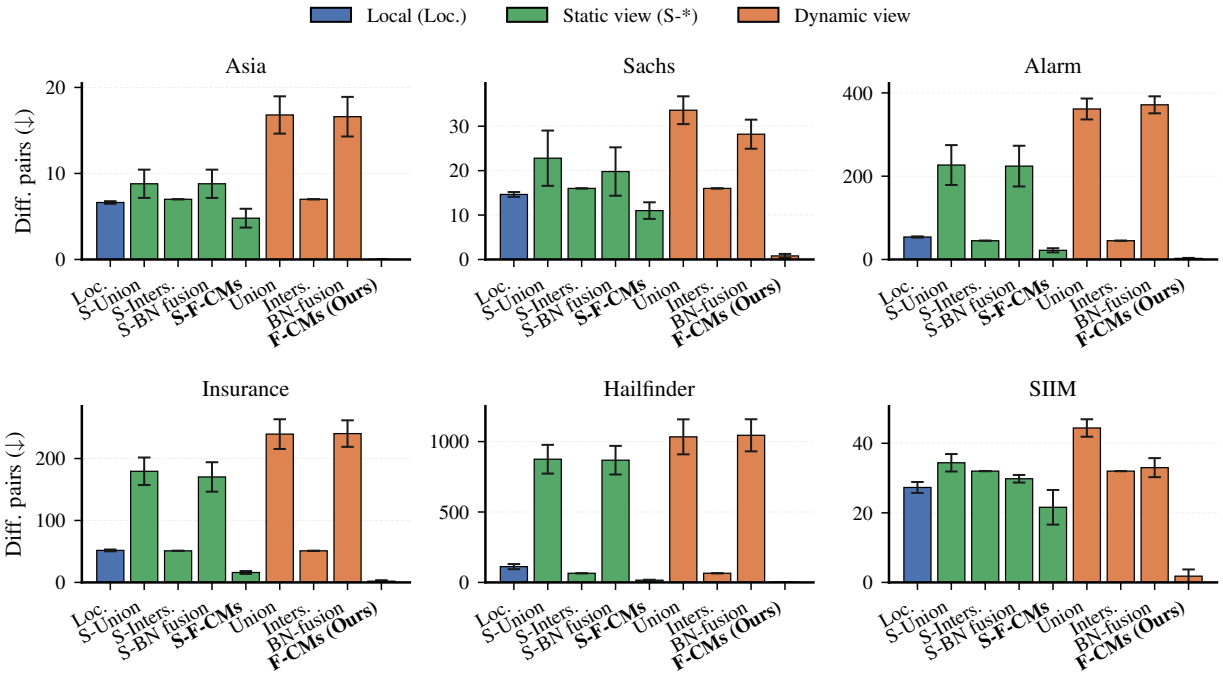


Figure 6: **Ablation of graph aggregation strategies under stronger perturbations.** *DiffPairs* (lower is better) for different aggregation rules under localized, static federated, and dynamic federated visibility, with graph alteration $p = 0.6$ and 60% corrupted clients, following the same experimental setup as Fig. 9 of the original manuscript.